Quantification: 543-2
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	12
interior lacuna count	12
frame-edge lacuna count	0

Interior lacunae only: mean, median and sample SD (n = 12)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	2027.92	2066	537.141	px^2
major_axis_length_px	lacuna shape	81.5016	86.4661	16.9465	px
minor_axis_length_px	lacuna shape	33.8846	32.3799	6.9212	px
aspect_ratio	lacuna shape	2.4727	2.3986	0.6666	unitless
eccentricity	lacuna shape	0.8942	0.9087	0.0558	unitless
solidity	lacuna shape	0.8887	0.9115	0.0668	unitless
roots_count	headline	7.6667	7.5	3.1718	count
ring_length_r30_px	headline	340.417	359.5	123.66	px
ring_length_r60_px	reported	984.417	931	246.946	px
owned_length_px	ownership-dependent	3155.58	2646.97	1753.29	px
edge_count	ownership-dependent	146.417	128.5	81.8518	count
mean_edge_length_px	ownership-dependent	21.9862	21.5468	3.0031	px
perimeter_px	normalised	222.101	229.693	45.067	px
ring_area_r30_px2	normalised	8742.17	8905.5	1127.78	px^2
ring_area_r60_px2	normalised	22189.9	23189	2825.36	px^2
in_frame_fraction_r30	normalised	0.9965	1	0.0123	unitless
in_frame_fraction_r60	normalised	0.9814	1	0.0596	unitless
ring_density_r30	normalised	0.0386731	0.0402737	0.0119805	px^-1
ring_density_r60	normalised	0.0443415	0.0461838	0.00876193	px^-1
roots_per_100px_perimeter	normalised	3.3796	3.7115	1.1821	per 100 px
ring_attached_length_r30_px	network v2	287.583	300.5	140.653	px
ring_attached_length_r60_px	network v2	794.5	712	325.929	px
ring_length_w_r30_px	network v2	384.535	401.2	143.337	px
ring_length_w_r60_px	network v2	1125.12	1075.15	287.232	px
sholl_crossings_r10	network v2	8.8333	10	3.6139	count
sholl_crossings_r20	network v2	11.8333	12	3.4333	count
sholl_crossings_r30	network v2	13.9167	13	4.1878	count
ring_width_mean_r30_px	width	7.5002	7.53	0.4024	px

Per field
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measure	value	units
analysed_area_px2	1.02424e+06	px^2
total_skeleton_length_px	40374	px
canalicular_length_density_per_px	0.0394185	px^-1
junction_count	1018	count
junction_density_per_px2	0.00099391	px^-2
lacuna_count	12	count
lacunae_per_px2	1.172e-05	px^-2
skeleton_component_count	155	count

mean_component_length_px	260.477	px
median_component_length_px	18	px
field_length_density_w_per_px	0.0449058	px^-1
field_density_without_flagged_per_px	0.0394185	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	465	px
unattached_component_count	26	count
canalicular_length_density_attached_per_px	0.0394185	px^-1
field_length_density_with_unattached_per_px	0.0398724	px^-1

Canalicular width, the new measure

measure	value	units
width_mean_px	8.0052	px
width_median_px	8	px
width_p10_px	5.6569	px
width_p90_px	10.198	px
width_pixels_used	40374	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel.
Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the
dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the
pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every
thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	2494	90.6112	36.7713	2.4642	0.914	0.9096	0.4535	88.1403	271.181	False
2	3004	86.0791	52.8366	1.6292	0.7895	0.772	0.3181	207.791	919.033	False
3	1698	63.4601	34.3535	1.8473	0.8408	0.955	0.5365	257.425	340.805	False
4	1631	87.5017	30.956	2.8266	0.9353	0.741	0.0248	297.032	38.7695	False
5	1204	55.3793	28.6478	1.9331	0.8558	0.9177	0.1008	394.263	364.613	False
6	2066	81.299	34.8468	2.333	0.9035	0.8619	0.1545	472.247	745.532	False
7	1233	57.0332	27.8479	2.048	0.8727	0.9278	0.052	459.387	484.83	False
8	2066	100.448	26.9826	3.7227	0.9632	0.9018	0.236	550.262	877.61	False
9	2656	108.838	32.5449	3.3442	0.9542	0.8877	-0.0333	642.811	193.327	False
10	2113	91.9731	30.2288	3.0426	0.9444	0.9135	0.0369	728.295	633.991	False
11	2173	86.8531	32.2149	2.6961	0.9287	0.9358	0.0768	787.166	480.067	False
12	1997	68.5432	38.3841	1.7857	0.8285	0.9402	0.1037	859.482	288.042	False

Network per lacuna

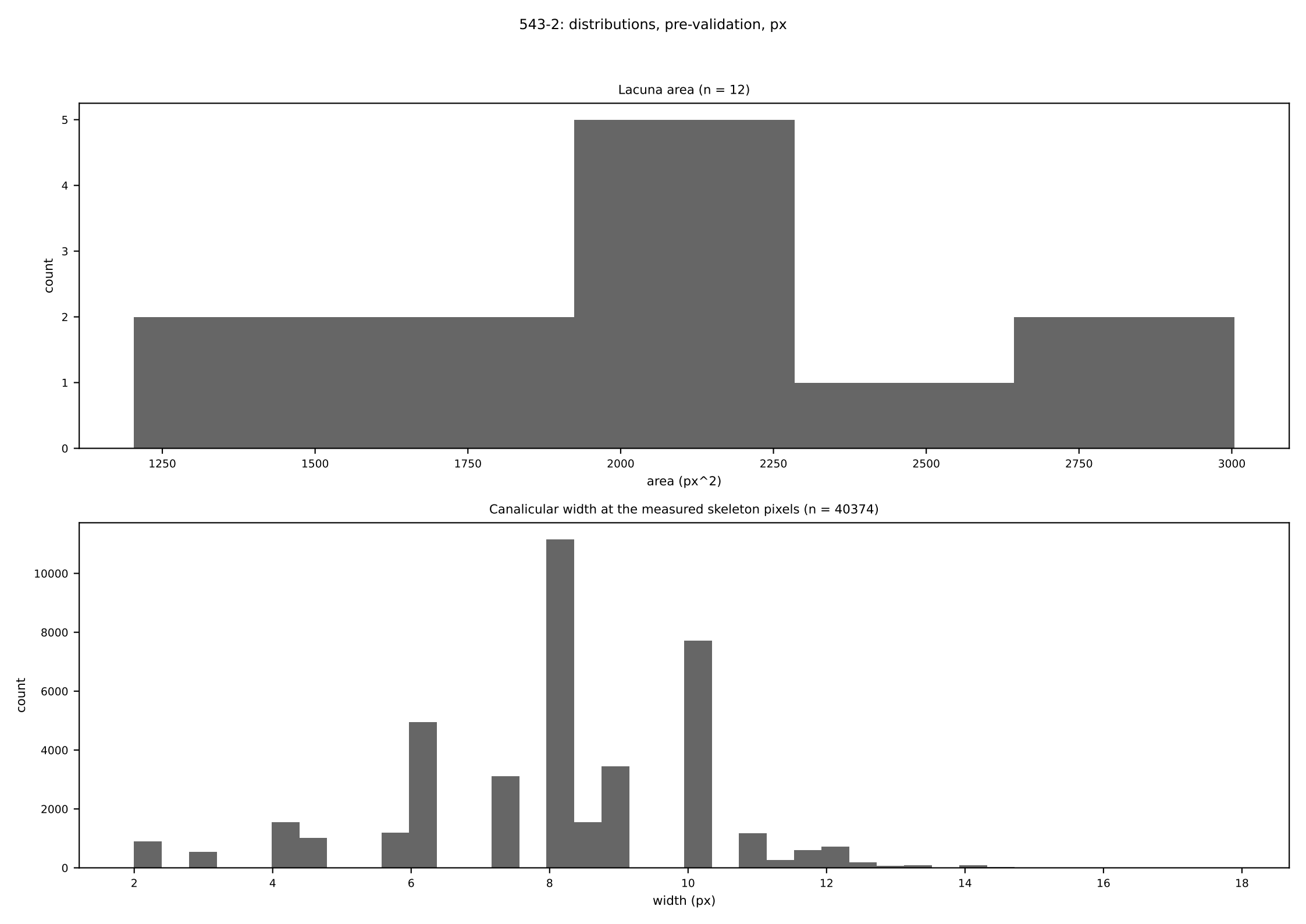
lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	11	435	1316	4215.58	193	21.8424	7.6346
2	13	627	1506	6168.75	295	20.911	7.2576
3	8	350	930	2010.74	90	22.3416	7.6209
4	10	365	691	1195.83	67	17.8482	6.9142
5	6	362	932	4654.03	219	21.2513	7.3821
6	9	366	1035	3670.02	152	24.1449	7.4392
7	1	133	746	719.085	24	29.9619	7.3183
8	9	373	1123	1938.37	99	19.5795	7.8895
9	7	231	783	2932.4	138	21.2492	7.8752
10	7	241	756	2208.05	96	23.0006	8.1534
11	7	357	1110	5792.59	265	21.8588	7.7386
12	4	245	885	2361.55	119	19.8449	6.7793

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	244.492	9510	23791	1	0.9843	0.0457413	0.055315	4.4991
2	295.12	10519	26271	1	1	0.0596064	0.0573256	4.405
3	172.894	7548	18977	1	1	0.0463699	0.0490067	4.6271
4	232.551	8756	18925	0.9575	0.7928	0.0416857	0.0365125	4.3001
5	154.267	7010	17818	1	1	0.0516405	0.0523067	3.8894
6	231.764	9080	22988	1	1	0.0403084	0.0450235	3.8833
7	155.924	7017	19168	1	1	0.018954	0.038919	0.6413
8	254.25	9429	23720	1	1	0.0395588	0.047344	3.5398
9	277.078	10094	25807	1	1	0.0228849	0.0303406	2.5264
10	227.622	8939	23523	1	1	0.0269605	0.0321388	3.0753
11	225.522	8872	23390	1	1	0.040239	0.0474562	3.1039
12	193.723	8132	21901	1	1	0.0301279	0.0404091	2.0648

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	366	1151	501.961	1522.47	10	16	16
2	620	1467	714.5	1720.17	16	19	21
3	314	757	394.291	1069.13	11	12	14
4	340	628	410.049	772.47	11	14	9
5	235	634	405.907	1081.18	5	12	11
6	347	1013	418.605	1187.66	11	11	11
7	62	317	150.711	851.725	4	7	9
8	345	937	430.919	1287.19	11	12	17
9	158	367	255.995	900.122	5	9	11
10	171	667	261.024	836.914	5	8	12
11	287	938	396.492	1252.97	10	13	21
12	206	658	273.966	1019.41	7	9	15


543-2: distributions, pre-validation, px
Lacuna area (n = 12)
count
area (px^2)
0
1
2
3
4
5
1250
1500
1750
2000
2250
2500
2750
3000
Canalicular width at the measured skeleton pixels (n = 40374)
count
width (px)
0
2000
4000
6000
8000
10000
2
4
6
8
10
12
14
16
18